SHARP LOWER BOUNDS FOR THE WILF NUMBER OF NUMERICAL SEMIGROUPS WITH SMALL DEFECT

RESEARCH REPORT

Abstract. Let S be a numerical semigroup with multiplicity m, embedding dimension e, conductor c, and left element count $L = |S \cap [0, F(S)]|$. The *Wilf number* of S is $W(S) = e \cdot L - c$. We define the *defect* $d = m - e$, which counts the decomposable elements of the Apéry set.

While Wilf's conjecture ($W(S) \geq 0$) remains open in general, Eliahou proved $W(S) \geq 0$ whenever $e \geq m/3$ (hence for all $d \leq 2m/3$). Our contribution is *quantitative*: we establish sharp linear lower bounds showing that the Wilf number grows with the multiplicity.

Theorem A. If $d = 1$ (i.e., $e = m - 1$), then $W(S) \geq m - 3$.

Theorem B. If $d = 2$ (i.e., $e = m - 2$), then $W(S) \geq 2m - 8$.

Both bounds are sharp: for each admissible m, there exist numerical semigroups attaining equality. We verify both theorems computationally on 258,582 numerical semigroups (genus ≤ 22) with zero violations, and propose a unified conjecture predicting $W_{\min}(m, d) = (m - d) \cdot \mathcal{L}(d) - 2m$ for all d, where $\mathcal{L}(d) = \lceil (\sqrt{8d + 1} - 1)/2 \rceil + 2$ is governed by the triangular numbers.

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1. Introduction

A *numerical semigroup* is a cofinite submonoid of $(\mathbb{N}, +)$, that is, a subset $S \subseteq \mathbb{N}$ containing 0, closed under addition, whose complement $\mathbb{N} \setminus S$ is finite. The cardinality $g(S) = |\mathbb{N} \setminus S|$ is the *genus*, the largest element of $\mathbb{N} \setminus S$ is the *Frobenius number* $F(S)$, and the *conductor* is $c(S) = F(S) + 1$. The smallest positive element $m(S) = \min(S \setminus \{0\})$ is the *multiplicity*, and the minimal number of generators is the *embedding dimension* $e(S)$.

Every numerical semigroup satisfies $1 \leq e \leq m$. When $e = m$, the semigroup has *maximal embedding dimension* (MED); semigroups with e close to m are said to have *nearly maximal embedding dimension*.

1.1. The Wilf number.

In 1978, Wilf [Wil78] observed that for $L(S) = |S \cap [0, F(S)]|$ (the number of "left elements"), the quantity

$$W(S) = e(S) \cdot L(S) - c(S) \tag{1}$$

appeared to be nonnegative. This led to:

Conjecture 1.1 (Wilf, 1978)**.** *For every numerical semigroup S, one has $W(S) \geq 0$.*

Despite substantial progress, Wilf's conjecture remains open in full generality. We refer to Delgado's surveys [Del20a, Del20b] for a comprehensive account.

1.2. Known results.

Sammartano [Sam12] proved the conjecture for $e \geq m/2$, i.e., $d \leq m/2$, and later Eliahou [Eli18, Eli20] extended this to $e \geq m/3$. In particular, Wilf's conjecture holds for all semigroups with $d \leq 2$. These proofs establish $W(S) \geq 0$ but do not provide explicit positive lower bounds depending on m.

Eliahou [Eli24] further investigated structural properties using the Kunz coordinates and the depth of the Apéry set.

1.3. Our contribution.

We introduce the *defect* $d = m - e$, which equals the number of decomposable elements in the Apéry set (see Section 2), and prove:

Theorem 1.2 (Theorem A)**.** *Let S be a numerical semigroup with multiplicity m and defect $d = 1$ (i.e., $e = m - 1$). Then*

$$W(S) \geq m - 3.$$

Moreover, this bound is sharp: for every $m \geq 3$, there exists a numerical semigroup with $d = 1$ achieving $W(S) = m - 3$.

Theorem 1.3 (Theorem B)**.** *Let S be a numerical semigroup with multiplicity $m \geq 4$ and defect $d = 2$ (i.e., $e = m - 2$). Then*

$$W(S) \geq 2m - 8.$$

Moreover, this bound is sharp: for every $m \geq 4$, there exists a numerical semigroup with $d = 2$ achieving $W(S) = 2m - 8$.

As a corollary, for $d = 1$ and $m \geq 4$, one has $W(S) \geq 1 > 0$; similarly, for $d = 2$ and $m \geq 5$, one has $W(S) \geq 2 > 0$. This provides strict positivity of the Wilf number in these families, contributing to a question raised by Moscariello and Sammartano.

Both theorems are verified computationally on 258,582 numerical semigroups of genus ≤ 22 with zero violations (Section 5).

Building on these results and extensive computational evidence, we formulate a unified conjecture (Section 6) predicting the exact minimum Wilf number $W_{\min}(m, d)$ for all defects d, governed by the triangular number function.

2. Preliminaries

2.1. **Apéry set and Kunz coordinates.** Let S be a numerical semigroup with multiplicity m. The *Apéry set* of S with respect to m is

$$\mathrm{Ap}(S, m) = \{w_0, w_1, \ldots, w_{m-1}\},$$

where w_i is the smallest element of S congruent to i modulo m, with $w_0 = 0$.

Definition 2.1 (Kunz coordinates)**.** For $i \in \{1, \ldots, m-1\}$, the *Kunz coordinate* (or *level*) of residue i is $k_i = w_i/m$. One has $k_i \geq 1$ for all $i \geq 1$.

Definition 2.2 (Depth)**.** The *depth* of S is

$$q(S) = k^* = \max\{k_i : 1 \leq i \leq m-1\}.$$

We denote by $r^* = \max\{i : k_i = k^*\}$ the largest residue attaining the maximum level.

The Frobenius number and conductor are expressed in Kunz coordinates as:

$$F(S) = k^* \cdot m + r^* - m = (k^* - 1)m + r^*, \qquad c(S) = (k^* - 1)m + r^* + 1. \tag{2}$$

2.2. **Decomposability and the defect.**

Definition 2.3. A residue $r \in \{1, \ldots, m-1\}$ is *decomposable* if there exist $a, b \in \{1, \ldots, m-1\}$ such that $a + b \equiv r \pmod{m}$ and

$$k_a + k_b + \varepsilon_{a,b} \leq k_r, \tag{3}$$

where $\varepsilon_{a,b} = \lfloor (a+b)/m \rfloor \in \{0, 1\}$.

It is a classical fact (see, e.g., [Eli20]) that the set of *primitive* (non-decomposable) residues is in bijection with the minimal generators of S other than m.

Definition 2.4 (Defect)**.** The *defect* of S is $d = m - e$, where e is the embedding dimension. Equivalently, d is the number of decomposable residues in $\{1, \ldots, m-1\}$.

Remark 2.5. Every residue r with $k_r = 1$ is automatically non-decomposable, since any pair (a, b) satisfies $k_a + k_b + \varepsilon_{a,b} \geq 1 + 1 + 0 = 2 > 1$.

2.3. **Left elements.** The number of left elements decomposes by residue class:

$$L = k^* + \sum_{i=1}^{m-1} \delta_i, \tag{4}$$

where δ_i is the *contribution* of residue i:

$$\delta_i = \begin{cases} k^* - k_i & \text{if } i \leq r^*, \\ \max(0, k^* - 1 - k_i) & \text{if } i > r^*. \end{cases} \tag{5}$$

Residue 0 contributes k^* elements (namely $0, m, 2m, \ldots, (k^* - 1)m$), while residue i contributes δ_i elements from its column in the Apéry table that lie below the Frobenius number. We denote the total residue contribution by $\Sigma\delta = \sum_{i=1}^{m-1} \delta_i$, so that $L = k^* + \Sigma\delta$.

2.4. **Sylvester's formula.** For a two-generator semigroup $S = \langle m, a\rangle$ with $\gcd(m, a) = 1$, the Frobenius number is $F = (m-1)(a-1) - 1$ and the Wilf number satisfies (see, e.g., [Syl84]):

$$W(S) = \frac{c(S)\cdot(m-3)}{2}. \tag{6}$$

In particular, $W(S) = 0$ when $m = 3$.

3. Theorem A: The case $d = 1$

Throughout this section, S denotes a numerical semigroup with multiplicity m and exactly one decomposable residue, i.e., $d = 1$ and $e = m-1$. We write r_d for the unique decomposable residue.

Theorem 3.1 (Theorem A)**.** *For every numerical semigroup S with $d = 1$,*

$$W(S) = (m-1)L - c \geq m-3.$$

Proof. The proof proceeds by a case analysis on L and k^*.

Case 1: $L = 3$. We first observe that $L \geq 3$ when $d = 1$. Indeed, $L = 1$ forces $e = m$ (MED), contradicting $d = 1$. If $L = 2$, then $\Sigma\delta = 2 - k^* \leq 0$ for $k^* \geq 2$, forcing all levels $\geq k^*$ (respectively $\geq k^* - 1$ for $i > r^*$), whence every pair sum $k_a + k_b + \varepsilon \geq 2 + 0 > 1$ exceeds any level, yielding $d = 0$. For $k^* = 1$ with $L = 2$, all levels equal 1 and $\Sigma\delta = 0$, giving $L = 1 \neq 2$. Thus $L \geq 3$.

We claim that $L = 3$ forces $k^* = 2$.

- If $k^* = 1$, then $L = 1 + 0 = 1 \neq 3$.
- If $k^* \geq 3$ and $L = 3$, then $\Sigma\delta = 3 - k^* \leq 0$, so all levels satisfy $k_i \geq k^* - 1 \geq 2$. Every pair sum is $\geq 2 + 2 = 4 > k^*$ for $k^* = 3$, and larger for $k^* > 3$. Thus $d = 0$, a contradiction.

So $k^* = 2$, and hence $c = m + r^* + 1 \leq m + (m-1) + 1 = 2m$, giving

$$W = (m-1)\cdot 3 - c \geq 3(m-1) - 2m = m-3.$$

Case 2: $L = 4$. We show that $k^* \in \{2, 3\}$. If $k^* \geq 5$, then $L \geq k^* \geq 5$, a contradiction. If $k^* = 4$, then $\Sigma\delta = 0$, forcing all levels ≥ 3; every pair sum is $\geq 6 > 4$, giving $d = 0$.

Subcase 2a: $k^* = 2$. Then $c \leq 2m$, and $W = (m-1)\cdot 4 - c \geq 4m - 4 - 2m = 2m - 4 \geq m - 3$.

Subcase 2b: $k^* = 3$. We need $c = 2m + r^* + 1$, and we claim $r^* \leq m - 2$. Suppose $r^* = m - 1$. Then all residues satisfy $i \leq r^*$, so $\Sigma\delta = \sum_{i=1}^{m-1}(3 - k_i) = 1$ (since $L = 4 = 3 + 1$). Thus exactly one residue j has $k_j = 2$, and all others have $k_i = 3$. For any pair (a, b): if both have level 3, the sum is $\geq 6 > 3$; if one is j (level 2) and the other has level 3, the sum is $\geq 5 > 3$; the self-sum (j, j) gives $4 + \varepsilon \geq 4 > 3$. So $d = 0$, a contradiction.

Therefore $r^* \leq m - 2$, giving $c \leq 2m + (m-2) + 1 = 3m - 1$, and

$$W = (m-1)\cdot 4 - c \geq 4m - 4 - (3m-1) = m-3.$$

Case 3: $L \geq 5$. We divide into subcases.

Subcase 3a: $m = 3$. Then $e = 2$ and $S = \langle 3, a\rangle$ for some a with $\gcd(3, a) = 1$. By (6), $W = 0 = m - 3$.

Subcase 3b: $m \geq 4$, $k^* \leq 3$. Then $c \leq (k^* - 1)m + (m-1) + 1 \leq 3m$, and

$$W \geq (m-1)\cdot 5 - 3m = 2m - 5 \geq m - 3.$$

Subcase 3c: $m \geq 4$, $k^* = 4$, $L = 5$. We claim this is impossible for $d = 1$. Since $L = 5$ and $k^* = 4$, we have $\Sigma\delta = 1$: exactly one residue j contributes $\delta_j = 1$, and all other residues contribute 0.

If $j \leq r^*$ with $k_j = 3$: all other levels are ≥ 3, so every pair sum is $\geq 3 + 3 = 6 > 4$, giving $d = 0$.

If $j > r^*$ with $k_j = 2$ (so $\delta_j = 4 - 1 - 2 = 1$): all levels are ≥ 3 except $k_j = 2$. For pairs not involving j: sum $\geq 6 > 4$. For (j, a) with $a \neq j$: sum $\geq 2 + 3 = 5 > 4$. For the self-sum (j, j): sum $= 4 + \varepsilon$. If $\varepsilon = 1$ (i.e., $2j \geq m$): sum $= 5 > 4$. If $\varepsilon = 0$ (i.e., $2j < m$): sum $= 4 = k^*$,

decomposing residue $2j$ with $k_{2j} = 4$. But $j > r^*$ implies $2j > 2r^* > r^*$, and $k_{2j} = 4 = k^*$ with $2j > r^*$ contradicts $r^* = \max\{i : k_i = k^*\}$.

Thus this configuration is impossible.

Subcase 3d: $m \geq 4$, $k^* = 4$, $L \geq 6$. Then $c \leq (k^* - 1)m + (m - 1) + 1 = 4m$, and

$$W \geq (m - 1) \cdot 6 - 4m = 2m - 6 \geq m - 3 \quad (m \geq 3).$$

Subcase 3e: $m \geq 4$, $k^* \geq 5$. This is the key case, resolved by the following lemma.

Lemma 3.2 (Pair Contribution Lemma)**.** *Let S be a numerical semigroup with $d = 1$ and $k^* \geq 5$. If the decomposable residue r_d admits a decomposition pair (a, b) with $k_a + k_b + \varepsilon = k_{r_d}$, then*

$$\delta_a + \delta_b \geq k^* - 2.$$

Proof. We have $k_a + k_b \leq k_{r_d} - \varepsilon \leq k^*$, and for $i > r^*$: $k_i \leq k^* - 1$.

Case (i): $a \leq r^*$ and $b \leq r^*$. Then $\delta_a + \delta_b = 2k^* - (k_a + k_b) \geq 2k^* - k^* = k^*$.

Case (ii): Exactly one of a, b is $\leq r^*$, say $a \leq r^*$ and $b > r^*$. Then $\delta_a + \delta_b = (k^* - k_a) + (k^* - 1 - k_b) = 2k^* - 1 - (k_a + k_b) \geq 2k^* - 1 - k^* = k^* - 1$.

Case (iii): $a > r^*$ and $b > r^*$. Then $\delta_a + \delta_b = 2(k^* - 1) - (k_a + k_b)$. If $\varepsilon = 0$: $k_a + k_b = k_{r_d} \leq k^*$, giving $\delta_a + \delta_b \geq 2k^* - 2 - k^* = k^* - 2$. If $\varepsilon = 1$: $k_a + k_b = k_{r_d} - 1 \leq k^* - 1$, giving $\delta_a + \delta_b \geq 2k^* - 2 - (k^* - 1) = k^* - 1$.

In all cases, $\delta_a + \delta_b \geq k^* - 2$. □

We now use the Pair Contribution Lemma to complete the proof. Note that $c = (k^* - 1)m + r^* + 1 \leq k^* m$.

When $a \neq b$: Since a and b are distinct, $\Sigma\delta \geq \delta_a + \delta_b \geq k^* - 2$, so $L \geq k^* + (k^* - 2) = 2k^* - 2$. Then

$$W \geq (m - 1)(2k^* - 2) - k^* m = k^*(m - 2) - 2(m - 1).$$

For $k^* \geq 5$: $W \geq 5(m - 2) - 2(m - 1) = 3m - 8 \geq m - 3$.

When $a = b$ (self-sum): The residue a appears only once in $\Sigma\delta$, so we only have $\Sigma\delta \geq \delta_a$. From the decomposition condition $2k_a + \varepsilon \leq k^*$, we get $k_a \leq \lfloor k^*/2 \rfloor$. We subdivide further.

(i) $a \leq r^*$: $\delta_a = k^* - k_a \geq k^* - \lfloor k^*/2 \rfloor = \lceil k^*/2 \rceil \geq 3$. So $L \geq k^* + \lceil k^*/2 \rceil$. Using $\lceil k^*/2 \rceil \geq k^*/2$:

$$W \geq (m - 1) \cdot \tfrac{3k^*}{2} - k^* m = \frac{k^*(m - 3)}{2}.$$

For $k^* \geq 5$ and $m \geq 4$: $W \geq 5(m - 3)/2 \geq m - 3$.

(ii) $a > r^*$ and $m = 4$: Since $a > r^* \geq 1$ and $a \leq m - 1 = 3$, we have $2a \geq 4 = m$, so $\varepsilon = 1$, giving $k_a \leq \lfloor (k^* - 1)/2 \rfloor$ and $\delta_a = k^* - 1 - k_a \geq \lceil (k^* - 1)/2 \rceil$. Then $L \geq k^* + \lceil (k^* - 1)/2 \rceil$ and

$$W \geq 3\big(k^* + \lceil (k^* - 1)/2 \rceil\big) - 4k^* = 3\lceil (k^* - 1)/2 \rceil - k^*.$$

For $k^* = 5$: $W \geq 3 \cdot 2 - 5 = 1 = m - 3$. For $k^* \geq 6$: the expression is nondecreasing and ≥ 2.

(iii) $a > r^*$ and $m \geq 5$: $\delta_a = k^* - 1 - k_a \geq \lceil k^*/2 \rceil - 1 \geq 2$. So $L \geq k^* + 2$.

If $k^* \leq m + 1$: $W \geq (m - 1)(k^* + 2) - k^* m = 2(m - 1) - k^* \geq 2(m - 1) - (m + 1) = m - 3$.

If $k^* \geq m + 2$: using $\delta_a \geq (k^* - 2)/2$, we get $L \geq k^* + (k^* - 2)/2 = (3k^* - 2)/2$, and

$$W \geq (m - 1) \cdot \frac{3k^* - 2}{2} - k^* m = \frac{k^*(m - 3) - 2(m - 1)}{2}.$$

It suffices to check $k^*(m - 3) \geq 4(m - 2)$, i.e., $k^* \geq 4(m - 2)/(m - 3)$. For $m = 5$: need $k^* \geq 6$, and indeed $k^* \geq m + 2 = 7$. For $m \geq 6$: $4(m - 2)/(m - 3) < 6 \leq k^*$.

In all subcases, $W \geq m - 3$. □

3.1. **Sharpness.**

Proposition 3.3. *For every $m \geq 3$, the bound $W \geq m-3$ is attained.*

Proof. For $m = 3$: $S = \langle 3, a\rangle$ has $e = 2$, $d = 1$, and $W = 0 = m - 3$ by (6).

For $m \geq 4$, consider the family T_m of semigroups with $k^* = 2$ and $r^* = m-1$, yielding $c = 2m$, $L = 3$ (with $\Sigma\delta = 1$, coming from a single level-1 residue that is decomposable). Then $W = (m-1)\cdot 3 - 2m = m - 3$.

Explicitly, $T_m = \langle m, m+1, 2m+3, 2m+4, \ldots, 3m-1\rangle$ has $e = m-1$, $c = 2m$, $L = 3$, and $W = m - 3$. □

Corollary 3.4. *For $d = 1$ and $m \geq 4$, the Wilf number is strictly positive: $W(S) > 0$.*

4. Theorem B: The case $d = 2$

Throughout this section, S denotes a numerical semigroup with multiplicity m and defect $d = 2$, so that $e = m-2$ and $W(S) = (m-2)L - c$.

Theorem 4.1 (Theorem B)**.** *For every numerical semigroup S with $d = 2$ and $m \geq 4$,*

$$W(S) = (m-2)L - c \geq 2m - 8.$$

The proof is divided into cases according to k^*.

4.1. **Case $k^* = 2$.**

Lemma 4.2. *If $d = 2$ and $k^* = 2$, then $L \geq 4$.*

Proof. We have $L = 2 + \Sigma\delta$. Suppose $L \leq 3$, so $\Sigma\delta \leq 1$.

For $k^* = 2$: residues $i > r^*$ are forced to have $k_i = 1$ (since $1 \leq k_i \leq k^* - 1 = 1$). Residues $i \leq r^*$ have $k_i \in \{1, 2\}$, with at most one at level 1 (since $\Sigma\delta \leq 1$).

A residue $r \leq r^*$ is decomposable if there exist a, b with $k_a = k_b = 1$, $\varepsilon = 0$ (i.e., $a + b = r < m$), and $k_r = 2$. The level-1 residues are $\{r^*+1, \ldots, m-1\} \cup N_1$ where $N_1 \subseteq \{1, \ldots, r^*\}$ with $|N_1| \leq 1$.

For any pair (a, b) with $a, b > r^*$: $a + b \geq 2(r^*+1) > r^*$, so $a + b$ does not yield a residue $\leq r^*$ (without a carry). Mixed pairs ($a \in N_1, b > r^*$): $a + b > r^*$. Self-sum from N_1: at most one decomposition.

Thus $d \leq 1$, contradicting $d = 2$. □

With $L \geq 4$ and $c = m + r^* + 1 \leq 2m$:

$$W \geq (m-2)\cdot 4 - 2m = 2m - 8.$$

4.2. **Case $k^* = 3$.**

Lemma 4.3. *If $d = 2$ and $k^* = 3$, then $L \geq 5$.*

Proof. Suppose $L = 4$, so $\Sigma\delta = 1$. Either a single residue $j \leq r^*$ has $k_j = 2$ (all others ≥ 2), or a single residue $j > r^*$ has $k_j = 1$ (all $i \leq r^*$ have $k_i = 3$, all other $i > r^*$ have $k_i = 2$).

In the first subcase, all levels are ≥ 2, so any pair sum is $\geq 2 + 2 = 4 > 3 = k^*$, giving $d = 0$.

In the second subcase, the only pair producing a sum ≤ 3 involves j: the self-sum (j, j) gives $1 + 1 + \varepsilon \leq 3$, producing at most one decomposition. Any other pair (j, a) with $k_a \geq 2$ gives sum $\geq 3 + \varepsilon$; if $a > r^*$ and $j + a < m$, the target has level $\leq 2 < 3$, which fails. If $a \leq r^*$, $k_a = 3$ gives sum $\geq 4 > 3$.

Thus $d \leq 1$ in both subcases, a contradiction. □

Lemma 4.4. *If $d = 2$, $k^* = 3$, and $L = 5$, then $r^* \leq m - 3$.*

Proof. We show $r^* = m-1$ and $r^* = m-2$ are both impossible.

$r^* = m-1$*:* Then $\Sigma\delta = 2$. Either two residues have level 2 (all levels ≥ 2, pair sums $\geq 4 > 3$, $d = 0$), or one residue has level 1 (self-sum gives at most one decomposition, $d \leq 1$).

$r^* = m-2$*:* Then residue $m-1$ has $k_{m-1} \leq 2$, and $\Sigma\delta = 2$. Careful analysis shows the self-sum $(m-1, m-1)$ gives $1+1+1 = 3$ (since $2(m-1) \geq m$), decomposing residue $m-2 = r^*$ with $k_{r^*} \leq 3$. This produces at most one decomposition. All other pairs involving residue $m-1$ give $k_{m-1} + k_a + 1 \geq 1+2+1 = 4 > 3$. Thus $d \leq 1$. □

With $L = 5$ and $r^* \leq m-3$: $c = 2m + r^* + 1 \leq 2m + (m-3) + 1 = 3m-2$, giving

$$W = (m-2)\cdot 5 - c \geq 5m - 10 - (3m-2) = 2m-8.$$

For $L \geq 6$: $c \leq 2m + (m-1) + 1 = 3m$, so $W \geq (m-2)\cdot 6 - 3m = 3m - 12 \geq 2m - 8$ for $m \geq 4$.

4.3. **Case $k^* \geq 4$.**

Lemma 4.5 (Deficit Lemma)**.** *If $d = 2$ and $k^* \geq 4$, then $\Sigma\delta \geq k^* - 1$, i.e., $L \geq 2k^* - 1$.*

Proof. We prove the contrapositive: if $\Sigma\delta \leq k^* - 2$, then $d \leq 1$.

Each decomposition of a residue by a pair (a, b) "consumes" a certain amount of deficit. We analyze three subcases based on the positions of a and b relative to r^*.

Subcase A: $a \leq r^*$ and $b \leq r^*$. The consumed deficit is $\delta_a + \delta_b = 2k^* - (k_a + k_b) \geq k^*$. A single such pair exhausts a budget of $k^* > k^* - 2$. Thus no such pair can arise when $\Sigma\delta \leq k^* - 2$.

Subcase B: One of a, b is $\leq r^*$, the other $> r^*$. The deficit consumed is $\geq k^* - 1$. After one such pair, the remaining budget is $\leq (k^* - 2) - (k^* - 1) = -1 < 0$. No second decomposition pair can exist.

Subcase C: $a > r^*$ and $b > r^*$. The deficit consumed is $\geq k^* - 2$. After one pair, the remaining budget is ≤ 0. If a second pair shares an element a with the first, the combined deficit of the three residues $\{a, b_1, b_2\}$ is $\geq k^* - 3 + k_a \geq k^* - 2$ (since $k_a \geq 1$). For $k_a = 1$, we have $a > r^*$ and $k_{b_j} \leq k^* - 2$, giving $\delta_{b_j} \geq 1$, so the total deficit exceeds $k^* - 2$. If the second pair is disjoint from the first, it needs deficit $\geq k^* - 2$ on its own, giving a total $\geq 2(k^* - 2) > k^* - 2$ for $k^* \geq 4$.

For self-sums (a, a) in Subcase C: residue a contributes δ_a only once. A disjoint second pair requires deficit $\geq k^* - 2$, giving total $\geq \delta_a + (k^* - 2) \geq (k^* - 2)/2 + (k^* - 2) = 3(k^* - 2)/2 > k^* - 2$ for $k^* \geq 4$. An overlapping second pair sharing a reduces to the overlap case above.

In all subcases, $\Sigma\delta \leq k^* - 2$ allows at most one decomposition pair, hence $d \leq 1$. □

Completion of Theorem B for $k^* \geq 4$. We use $c \leq k^* m$ and $L \geq 2k^* - 1$ (Lemma 4.5).

Subcase $m \geq 6$:

$$W \geq (m-2)(2k^* - 1) - k^* m = k^*(m-4) - (m-2).$$

For $k^* \geq 4$ and $m \geq 6$: $(k^* - 2)(m-4) \geq 2 \cdot 2 = 4 \geq 2$, so $k^*(m-4) - (m-2) \geq 2(m-4) \geq 2m - 8$.

Subcase $m = 5$, $k^* \geq 5$: $W \geq 3(2k^* - 1) - 5k^* = k^* - 3 \geq 2 = 2 \cdot 5 - 8$.

Subcase $m = 5$, $k^* = 4$: Direct analysis of the four residues $\{1, 2, 3, 4\}$. For $d = 2$ with $k^* = 4$, two decomposition pairs require at least one residue with level ≤ 2. If a single residue j has $k_j = 1$, one decomposition arises from (j, j) and another from (j, b) with $k_b = 3$ and $j + b \equiv r^* \pmod 5$. The deficit analysis gives $L \geq 8$ and $c \leq 16 + r^* + 1$, whence $W = 3L - c \geq 24 - 16 - r^* \geq 24 - 20 = 4 \geq 2$.

Subcase $m = 4$: Then $e = 2$, so $S = \langle 4, g \rangle$ with $\gcd(4, g) = 1$. By (6), $W = c(4-3)/2 = c/2 \geq 0 = 2 \cdot 4 - 8$. □

4.4. Sharpness.

Proposition 4.6. *For every $m \geq 4$, the bound $W \geq 2m - 8$ is attained.*

Proof. For $m = 4$: $S = \langle 4, g \rangle$ with $d = 2$ gives $W = 0 = 2m - 8$.

For $m \geq 5$: Two families of tight semigroups exist.

(1) *Family with $k^* = 2$, $L = 4$, $c = 2m$, $r^* = m - 1$:* Kunz levels consist of $(m - 3)$ residues at level 2 and two residues at level 1. Then $W = (m - 2) \cdot 4 - 2m = 2m - 8$.

(2) *Family with $k^* = 3$, $L = 5$, $c = 3m - 2$, $r^* = m - 3$:* Kunz levels are all at 3 except two at level 1 (and one at level 2 for the decomposable residues). Then $W = (m-2)\cdot 5-(3m-2) = 2m - 8$.

Both families have been verified to exist for all m in the stated ranges. □

Corollary 4.7. *For $d = 2$ and $m \geq 5$, the Wilf number is strictly positive: $W(S) > 0$.*

5. Computational verification

We performed an exhaustive computational verification of Theorems A and B on the complete census of numerical semigroups of genus ≤ 22. The total number of such semigroups is 258,582, consistent with the values recorded in OEIS sequence A007323. The computation was completed in 7.2 seconds.

5.1. Summary of results.

Theorem	Semigroups tested	Violations	Status
A ($d = 1$, $W \geq m - 3$)	22,149	0	Verified
B ($d = 2$, $W \geq 2m - 8$)	39,159	0	Verified

5.2. Statistics for Theorem A ($d = 1$). Table 1 summarizes the minimum observed Wilf number by multiplicity. In all cases $W_{\min} \geq m - 3$, with equality achieved for $m \leq 12$. For $m \geq 13$, the observed minimum exceeds the bound $m - 3$; this is because the tight examples for large m have genus exceeding 22 and are not captured by the census.

Table 1. Theorem A verification: $d = 1$, genus ≤ 22.

m	Count	$W_{\min}$	$W_{\max}$	$\bar{W}$	$m - 3$	Tight
3	14	0	0	0.0	0	14
4	113	1	22	12.8	1	2
5	327	2	44	20.9	2	3
6	1,014	3	66	34.6	3	3
7	1,467	4	88	41.8	4	4
8	2,610	5	110	54.0	5	4
9	2,756	6	132	62.7	6	5
10	3,235	7	154	72.8	7	4
11	2,539	8	176	79.0	8	5
12	2,582	9	198	91.6	9	5

5.3. Statistics for Theorem B ($d = 2$).

TABLE 2. Theorem B verification: $d = 2$, genus ≤ 22.

m	Count	$W_{\min}$	$W_{\max}$	$\bar{W}$	$2m-8$	Tight
4	6	0	0	0.0	0	6
5	97	2	22	9.5	2	3
6	489	4	39	21.5	4	5
7	1,068	6	63	30.0	6	4
8	2,496	8	88	43.0	8	7
9	3,455	10	104	52.2	10	9
10	4,908	12	126	64.0	12	11
11	4,889	14	154	73.1	14	11
12	5,416	16	160	86.6	16	16
13	4,467	18	189	96.2	18	17

5.4. **Verification of key lemmas.** In addition to the main theorems, we verified each intermediate lemma:

Lemma	**Cases**	**Violations**
Pair Contribution (Lem. 3.2), $d = 1$, $k^* \geq 5$	1,537	0
$r^* \leq m-2$ for $d = 1$, $k^* = 3$, $L = 4$ (Case 2b)	55	0
$k^* = 4$, $L = 5$ impossible for $d = 1$ (Subcase 3c)	—	0 cases found
$L \geq 4$ for $d = 2$, $k^* = 2$ (Lem. 4.2)	3,648	0
$L \geq 5$ for $d = 2$, $k^* = 3$ (Lem. 4.3)	9,509	0
$r^* \leq m-3$ for $d = 2$, $k^* = 3$, $L = 5$ (Lem. 4.4)	90	0
Deficit Lemma (Lem. 4.5), $d = 2$, $k^* \geq 4$	4,443	0

5.5. **Tight examples.** For Theorem A, the tight examples ($W = m-3$) all share a common structure: $k^* = 2$, $r^* = m-1$, $c = 2m$, $L = 3$. An explicit tight example is $\langle 6, 11, 19, 20, 21\rangle$ with $W = 3$.

For Theorem B, tight examples ($W = 2m-8$) arise from two families: (i) $k^* = 2$, $L = 4$, $c = 2m$; and (ii) $k^* = 3$, $L = 5$, $c = 3m-2$. An example is $\langle 9, 16, 17, 28, 29, 30, 31\rangle$ with $W = 10 = 2 \cdot 9 - 8$.

6. A unified conjecture

The bounds of Theorems A and B fit into a larger pattern. Extensive computation for $d = 0, 1, \ldots, 15$ suggests the following:

Conjecture 6.1 (Unified sharp Wilf bound)**.** *For every numerical semigroup S with multiplicity m and defect $d = m - e$, and for m sufficiently large relative to d:*

$$W(S) \geq W_{\min}(m, d) = (m-d) \cdot \mathcal{L}(d) - 2m, \tag{7}$$

where

$$\mathcal{L}(d) = \left\lceil \frac{\sqrt{8d+1}-1}{2} \right\rceil + 2. \tag{8}$$

6.1. **The triangular number connection.** The function $\mathcal{L}(d)$ has a simple combinatorial interpretation. Let $p(d) = \lceil(\sqrt{8d+1}-1)/2\rceil$, the smallest integer p such that the triangular number $T_p = p(p+1)/2$ satisfies $T_p \geq d$. Then $\mathcal{L}(d) = p(d) + 2$.

Remark 6.2. The structural origin is as follows. The tight semigroups (those achieving $W = W_{\min}$) typically have $c = 2m$, with p level-1 and $(m-1-p)$ level-2 Apéry elements. Each decomposable residue (at level 2) must be the sum of two level-1 residues modulo m. With p elements at level 1,

the number of distinct pairwise sums (including self-sums) is at most $T_p = p(p+1)/2$. The condition $d \leq T_p$ forces $p \geq p(d)$, yielding $L = 2 + p \geq 2 + p(d) = \mathcal{L}(d)$.

6.2. **Predicted bounds.** Table 3 lists the predicted minimum Wilf number for each defect $d \leq 15$.

TABLE 3. Unified conjecture: predicted bounds for $d = 0, \ldots, 15$.

d	$p(d)$	$\mathcal{L}(d)$	Slope	$W_{\min}(m, d)$
0	0	2	0	0
1	1	3	1	$m - 3$
2	2	4	2	$2m - 8$
3	2	4	2	$2m - 12$
4	3	5	3	$3m - 20$
5	3	5	3	$3m - 25$
6	3	5	3	$3m - 30$
7	4	6	4	$4m - 42$
8	4	6	4	$4m - 48$
9	4	6	4	$4m - 54$
10	4	6	4	$4m - 60$
11	5	7	5	$5m - 77$
12	5	7	5	$5m - 84$
13	5	7	5	$5m - 91$
14	5	7	5	$5m - 98$
15	5	7	5	$5m - 105$

Remark 6.3. The slope of $W_{\min}$ with respect to m is $\mathcal{L}(d) - 2 = p(d)$, which increases by 1 each time d crosses a triangular number $T_k = k(k+1)/2$. The "constant" term is $-d \cdot \mathcal{L}(d)$.

6.3. **Computational evidence.** The conjecture has been verified on all 258,582 numerical semigroups of genus ≤ 22 for $d = 0, \ldots, 15$. In the stabilized regime (where m is large enough that the formula applies as a sharp bound), there are 66 exact matches out of 122 (d, m) pairs, with the remaining 56 cases satisfying $W_{\min}^{\text{obs}} > W_{\min}^{\text{pred}}$ (an artifact of the finite genus census, which does not capture the tightest examples at large m). There are zero violations.

For $d = 0$: the conjecture reduces to $W \geq 0$, the classical Wilf conjecture for MED semigroups, which is well known.

For $d = 1$ and $d = 2$: the conjecture yields our Theorems A and B.

For $d \geq 3$: further algebraic verifications of achiever families have been performed for $d = 3, \ldots, 15$ via targeted Kunz enumeration and explicit construction, confirming the predicted values.

7. CONCLUSION

We have established sharp quantitative lower bounds for the Wilf number of numerical semigroups with small defect: $W \geq m - 3$ for $d = 1$ and $W \geq 2m - 8$ for $d = 2$. These are, to our knowledge, the first results proving that the Wilf number grows linearly with the multiplicity in any infinite family of numerical semigroups, going beyond the qualitative statement $W \geq 0$.

Several directions for further investigation present themselves:

(1) **Higher defects.** Prove the unified conjecture for $d = 3$ (predicting $W \geq 2m - 12$) and beyond. The case structure becomes increasingly complex, but the Deficit Lemma technique (Lemma 4.5) may generalize.

(2) **Stabilization thresholds.** Determine the precise value of $m_{\min}(d)$ above which the formula $W_{\min}(m,d) = (m-d)\cdot\mathcal{L}(d) - 2m$ holds as a sharp bound. Boundary cases (where $d = T_{p(d)}$ exactly) require a Sidon-like condition on the level-1 residues, leading to higher stabilization thresholds.

(3) **Structural characterization.** Classify all tight semigroups (those achieving the minimum Wilf number) for each defect. For $d = 1$, tight semigroups are characterized by $k^* = 2$, $c = 2m$, $L = 3$ (for $m \geq 4$).

(4) **Connection to Eliahou's graph.** Explore whether the deficit budget approach can be unified with Eliahou's graph-theoretic framework [Eli20] for studying the Wilf number.

(5) **Asymptotics.** The slope $p(d) \sim \sqrt{2d}$ grows sublinearly in d. Understanding the behavior of $W_{\min}$ as $d \to \infty$ (with $d/m \to \alpha$ for fixed $\alpha < 2/3$) would provide insight into the "difficulty frontier" for Wilf's conjecture.

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